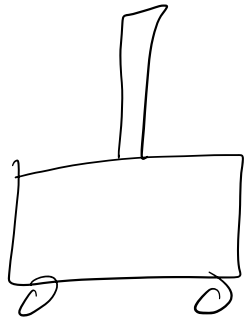

- M_c - Mass of the cart/main Body
- M_p - Mass of the inverted Pendulum
- L - Length from the Pivot to Pendulum Center of Mass
- I - Inertia of the Pendulum
- r - Radius of the wheel

$\theta(t)$ - Angle of the Pendulum ($\theta = 0$, Perfectly upright)

$x(t)$ - Horizontal Position of the cart

<u>Energy Analysis</u> (Lagrangian Mechanics)

$$L = T - V = \left(\begin{array}{c} \text{Kinetic} \\ \text{Energy} \end{array} - \begin{array}{c} \text{Potential} \\ \text{Energy} \end{array} \right)$$

<u>Euler - Lagrange Method</u>:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \left(\frac{\partial L}{q_i}\right) = Q_i$$

q_i = Position variable

$\dot{q}_i$ = Velocity variable

Kinetic Energy

$T_{cart} = \frac{1}{2} M_c \dot{x}^2$ - Cart Energy

V_p - Velocity of the Pendulums center of mass

Horizontal Velocity : $V_{px} = \dot{x} + L\dot{\theta}\cos(\theta)$

Vertical Velocity : $V_{py} = L\dot{\theta}\sin(\theta)$

$\Rightarrow V_p^2 = V_{px}^2 + V_{py}^2 = \dot{x}^2 + 2L\dot{x}\dot{\theta}\cos(\theta) + L^2\dot{\theta}^2$

$\Rightarrow T_{Pendulum} = \frac{1}{2} M_p (V_p^2) + \frac{1}{2} I \dot{\theta}^2$

Translational + Rotational

Total Kenitic Energy :

$$T_{total} = \frac{1}{2}(M_c + M_p)\dot{x}^2 + M_p L \dot{x}\dot{\theta}\cos(\theta) + \frac{1}{2}(M_p L^2 + I)\dot{\theta}^2$$

Potential Energy :

$$V = M_p g L \cos(\theta)$$

- The Lagrangian (L)

$$L = T - V$$

$$L = \frac{1}{2}(M_c + M_p)\dot{x}^2 + M_p L \dot{x}\dot{\theta}\cos(\theta) + \frac{1}{2}(M_p L^2 + I)\dot{\theta}^2 - M_p L \cos(\theta)$$

Equations of Motion : Euler - Lagrange

Cart:

$$\frac{\partial L}{\partial \dot{x}} = (M_c + M_p)\dot{x} + M_p L \dot{\theta} \cos(\theta)$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = (M_c + M_p)\ddot{x} + M_p L \ddot{\theta} \cos\theta - M_p L \dot{\theta}^2 \sin\theta$$

$$\frac{\partial L}{\partial x} = 0 \qquad F = \frac{\tau}{r}$$

τ ← Motor Torque

r ← Wheel Radius

$$\Rightarrow (M_c + M_p)\ddot{x} + M_p L \ddot{\theta} \cos(\theta) - M_p L \dot{\theta}^2 \sin(\theta) = \frac{\tau}{r}$$

Pendulum

$$\frac{\partial L}{\partial \dot{\theta}} = M_p L \dot{x} \cos(\theta) + (M_p L^2 + I)\dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = M_p L \ddot{x} \cos(\theta) - M_p L \ddot{x} \dot{\theta} \sin(\theta) + (M_p L^2 + I)\ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = -M_p L \dot{x} \dot{\theta} \sin(\theta) + M_p g L \sin(\theta)$$

NON Linear Eq of Motion for Pendulum

$$(M_p L^2 + I)\ddot{\theta} + (M_p L \ddot{x} \cos(\theta)) - M_p g L \sin(\theta) = 0$$

Linearization of non linear system

- Small Angle Approximation ($\theta \approx 0$)
  - $\cos(\theta) \approx 1$
  - $\sin(\theta) \approx 0$
  - $\dot{\theta}^2 \approx 0$

$$\Rightarrow \quad \begin{aligned} &1)\ (M_c + M_p)\ddot{x} + M_p L \ddot{\theta} = \frac{\tau}{r} \\ &2)\ (M_p L^2 + I)\ddot{\theta} + M_p L \ddot{x} = M_p g L \theta \end{aligned}$$

State Space Representation:

Acceleration Eq:

$$\ddot{x} = \frac{M_p^2 g L^2}{I(M_c + M_p) + M_c M_p L^2}\theta + \frac{M_p L^2 + I}{r[I(M_c + M_p) + M_c M_p L^2]}\tau$$

$$\ddot{\theta} = \frac{M_p g L (M_c + M_p)}{I(M_c + M_p) + M_c M_p L^2}\theta + \frac{M_p L}{r[I(M_c + M_p) + M_c M_p L^2]}\tau$$

$$Den = I(M_c + M_p) + M_c M_p L^2$$

$$\dot{x} = Ax + Bu$$

$x = [x, \dot{x}, \theta, \dot{\theta}]^T$ – State space form

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{-M_p^2 g L^2}{Den} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{M_p g L (M_c + M_p)}{Den} & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{M_p L^2 + I}{(Den) r} \\ 0 \\ \frac{-M_p L}{(r)(Den)} \end{bmatrix} \tau$$

LQR Controller Design

K (LQR Gains) stabalizes the system

The cost Function (J)

$$J = \int_0^\infty (x^T Q x + u^T R u)\, dt$$

x - State Error
u - Control Input
Q - State Cost Matrix (Penalty on Error)
R - Input Cost Matrix (Penalty on Effort)

$$U = -Kx$$

Use Algebraic Riccati Equation (ARE)

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

- solve for P via matlab

Then

$$K = R^{-1} B^T P$$

PID

Outerloop

$$\ddot{x} = g\theta$$

$$s^2 x(s) = g\,\theta(s)$$

Transfer function $= \dfrac{x(s)}{\theta(s)} = \dfrac{g}{s^2}$

Goals

1) T_s (settling time) $= 5\,s$

2) ζ (Damping ratio) $= 0.9$ (critical Damping)

Controller Math

PD eq

$$U(s) = (K_p + K_d s)\,E(s)$$

Characteristic Equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$1 + C(s)G(s) = 0$$

$$\Rightarrow \quad 1 + (K_p + K_d s)\frac{g}{s^2} = 0$$

$$\Rightarrow \quad s^2 + (K_p + K_d s)\,g = 0$$

$$\Rightarrow \quad s^2 + (g K_d)s + g K_p = 0$$

Natural freq (ω_n): $T_s = \dfrac{4}{\zeta\omega_n}$

$$5 = \frac{4}{(0.9)(\omega_n)} \quad \Rightarrow \quad \underline{\omega_n = 0.88 \text{ rad/s}}$$

Proportional Gain (K_p): $g \cdot K_p = \omega_n^2$

$$\Rightarrow (9.81)(K_p) = (0.88)^2 \Rightarrow \underline{K_p \approx 0.08}$$

Derivative Gain (K_d): $g K_d = 2 \zeta \omega_n$

$$\Rightarrow (9.81)(K_d) = 2(0.9)(0.88)$$

$$\Rightarrow \underline{K_d = 0.16}$$

Integral Gain (K_i): $\frac{\omega_n}{50} = \frac{0.88}{50} = \underline{0.017}$